

Safer Ground, Tighter Markets

A Dynamic Two-Market Model of Housing Adjustment under Risk Rating 2.0

Kosar Hosseini · Katherine Yoder Zipp · Scott Colby · Zhengqi Liu
Pennsylvania State University · SUNY Morrisville

THE RIVER IS THE ARGUMENT

- **01** Why the law **02** The damage **03** The reform
← **04** Two banks **05** The model
→ **06** The numbers **07** The recommendation



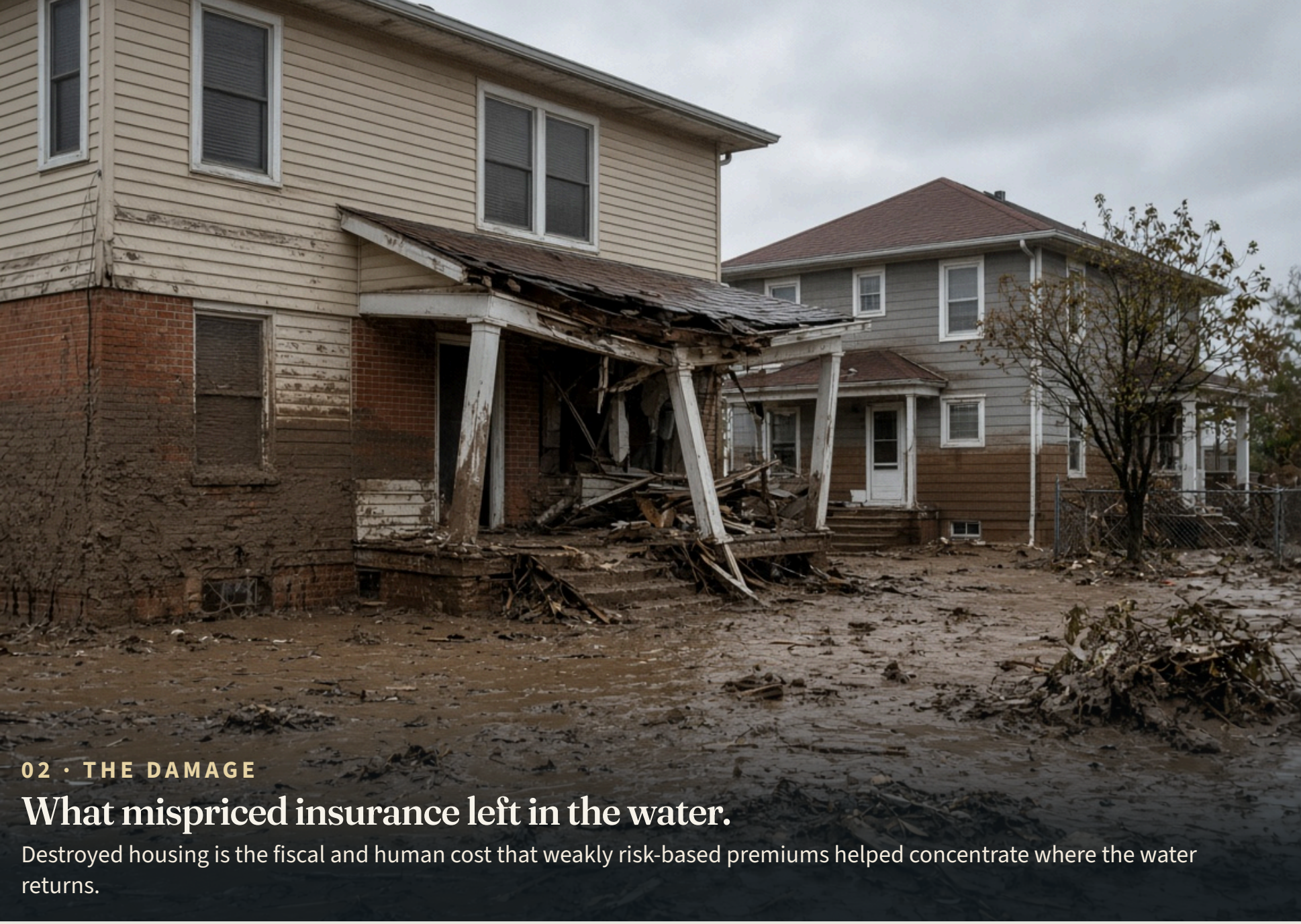
01 · MOTIVATION

Why price the floodplain at all?

Flooding is among the most common and costly natural hazards in the United States. It generates billions of dollars in damages and large fiscal and housing-market externalities.

For decades, subsidized or weakly risk-based National Flood Insurance Program premiums lowered the user cost of high-risk housing and encouraged development on exposed ground. Taxpayers and the government bore that mispricing.

FEMA's Risk Rating 2.0 aligns premiums with property-level flood risk. Raising the cost of flood-prone housing is a persistent user-cost shock. It can move prices and stocks inside the floodplain and in nearby safer markets. Hedonic studies capitalize flood risk into prices, but they hold the housing stock fixed, rarely model spillovers to safer markets, and do not separate short-run capitalization from long-run quantity adjustment.



02 · THE DAMAGE

What mispriced insurance left in the water.

Destroyed housing is the fiscal and human cost that weakly risk-based premiums helped concentrate where the water returns.

03 · THE INSURER

An expected 82% user-cost shock

82%

Expected NFIP premium increase in Allegheny County, the baseline calibration.

18%

The annual cap FEMA generally applies. It is not what forward-looking households capitalize.

$$\Psi_{FP}(m) = \Psi_{FP}^0 + f \cdot m \quad f = 0.0579$$

Households internalize the full expected increase at once. Ψ_{NF} does not change, so safer markets move only through demand.



THE QUESTION

Does the law only reprice the floodplain — or does it also tighten safer ground?

05 · THE MODEL · READ AFTER THE TWO BANKS

Prices jump. Then the stock moves.

A dynamic asset-price system after Poterba (1984) and Colby & Zipp (2021). Supply elasticity draws on Saiz (2010). Market 1 is non-floodplain; market 2 is floodplain.

DEMAND · GORMAN POLAR

$$R(S_{NF} + S_{FP}) = \frac{\alpha(W - \beta_L)}{(S_{NF} + S_{FP}) - \beta_S(1 - \alpha)}$$

SUPPLY · HOLMES & SIEG (2015)

$$Y_i(P_i) = a_i P_i^f$$

$$a_i = \frac{\delta S_i}{P_i^f}$$

$$i \in \{1, 2\}$$

$$\dot{P}_{NF}(t) = \Psi_{NF} P_{NF}(t) - R(S_{NF}(t) + S_{FP}(t))$$

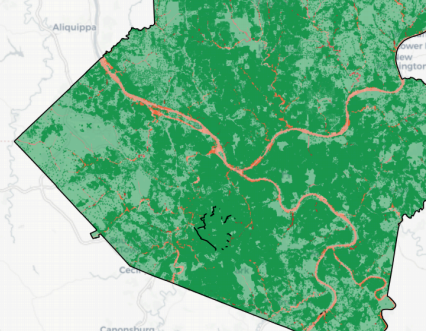
$$\dot{P}_{FP}(t) = \Psi_{FP}(m) P_{FP}(t) - R(S_{NF}(t) + S_{FP}(t))$$

$$\dot{S}_{NF}(t) = a_{NF} P_{NF}(t)^f - \delta S_{NF}(t)$$

$$\dot{S}_{FP}(t) = a_{FP} P_{FP}(t)^f - \delta S_{FP}(t)$$

$$\Psi_{FP}(m) = \Psi_{FP}^0 + f \cdot m$$

On impact the stock is predetermined, so the user-cost shock is capitalized into prices: a jump to the stable saddle path. Over time, exits, depreciation, redevelopment, and new construction change the stock. That is the separation between short-run capitalization and long-run spatial reallocation.



Housing stock

Median price

Market share

W \$76,000 ACS 2022

ϵ_w 0.08 short run

θ 0.26 BLS CEL 2022

Ψ_{NF} 0.059

ϵ_w 1.20 Saiz 2010

α 43.446

Ψ_{FP}^0 0.1169

δ 0.02 Harding et al.

β_L -56,716

f 0.0579 premium

ϵ_{FW} 0.67 Albouy et al.

β_S 1.50

04 · MECHANISM · THE CURRENT ARRIVES HERE

One shock. Two banks.

Inside the floodplain

1. User cost rises: $\Psi_{FP} \uparrow$
2. Willingness to pay falls
3. Price jumps down on impact: $P_{FP} \downarrow$
4. Construction and redevelopment slow
5. Long run: the stock S_{FP} contracts

Outside the floodplain

1. Demand shifts toward safer housing
2. Prices come under pressure: $P_{NF} \uparrow$
3. Construction responds
4. Long run: the stock S_{NF} expands
5. The outside move is only indirect

Safer-market expansion only partly offsets the floodplain contraction, so total regional stock falls.

No arbitrage. Free mobility. Perfect substitution between floodplain and non-floodplain housing in the baseline. Because Ψ_{NF} is unchanged, the outside market does not fully replace what leaves the floodplain.

← THE CURRENT TURNS

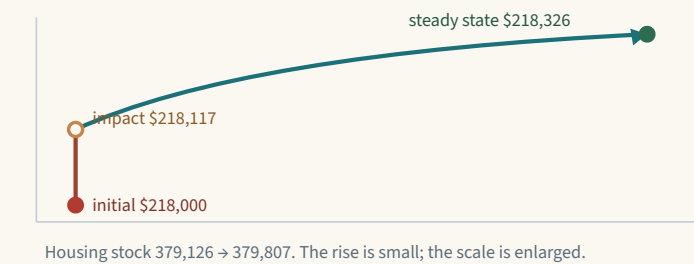
06 · EVIDENCE · ALLEGHENY COUNTY, 82% PREMIUM INCREASE

The floodplain takes the loss. Safer ground barely moves.

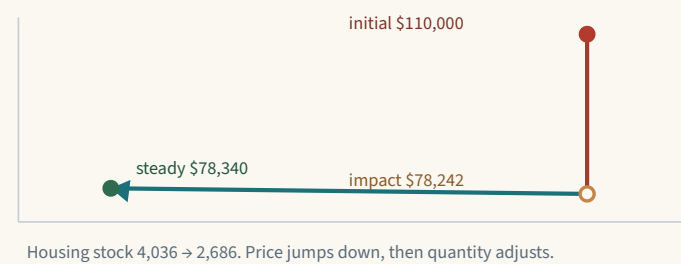
On impact, stocks are fixed. Floodplain prices fall 28.871%, and safer prices rise 0.054%. In the long run the floodplain stock falls 33.456%, the safer stock rises 0.180%, and total regional stock falls 0.175%. The safer market is 98.95% of the initial stock, so the spillover is positive and small — and it does not replace the housing that leaves.

	-29% Floodplain price, short run Exact -28.871% · \$110,000 → \$78,242. Long run -28.782%.	-33% Floodplain stock, long run Exact -33.456% · 4,036 → 2,686. Short run: no change.	+0.15% Safer price, long run Exact +0.150% · \$218,000 → \$218,326. Impact +0.054%.	+0.18% Safer stock, long run Exact +0.180% · 379,126 → 379,807.	-0.17% Total regional stock Exact -0.175% · 383,162 → 382,493.
OUTCOME					
Non-floodplain price	\$218,000		\$218,117 (+0.054%)	\$218,326 (+0.150%)	\$218,326 (+0.150%)
Floodplain price	\$110,000		\$78,242 (-28.871%)	\$78,242 (-28.782%)	\$78,340 (-28.782%)
Non-floodplain stock	379,126 (98.95%)		379,126 (98.95%)	379,807 (+0.180%)	379,807 (+0.180%)
Floodplain stock	4,036		4,036 (0%)	2,686 (-33.456%)	2,686 (-33.456%)
Total regional stock	383,162		383,162 (0%)	382,493 (-0.175%)	382,493 (-0.175%)

MARKET 1 · OUTSIDE THE FLOODPLAIN · AXES ZOOMED



MARKET 2 · INSIDE THE FLOODPLAIN · AXES ZOOMED



Miami-Dade · when flood risk covers half the county, safer ground moves

About 50.48% of residential properties are in A or V. Unlike Allegheny, risk is not a localized vein. Point estimates for an 85% premium increase; short-run stocks are fixed.

MARKET	PROPERTIES	PREMIUM RATE	SHORT-RUN PRICE	LONG-RUN PRICE	LONG-RUN STOCK
X · outside SFHA	454,117	—	+1.04%	+2.60%	+3.13%
A · inland SFHA	461,477	0.64%	-6.79%	-5.25%	-6.26%
V · coastal high-hazard	4,633	1.30%	-17.94%	-16.68%	-19.67%

Across premium increases from 15% to 100%: short-run prices X +0.20 to +1.21%, A -1.26 to -7.92%, V -3.73 to -20.52%. Long-run prices X +0.48 to +3.03%, A -0.97 to -6.12%, V -3.48 to -19.07%. Long-run stocks X +0.58 to +3.65%, A -1.16 to -7.30%, V -4.17 to -22.43%.

$$\dot{P}_i = \psi_i(s) P_i - R(S_X + S_A + S_V) \quad \dot{S}_i = a_i P_i^f - \delta S_i \quad i \in \{X, A, V\}$$

CES housing services. Let $a = (\sigma - 1) / \sigma$ and $\gamma = 0.989467$, the non-floodplain share. $H_i = [\gamma S_{NF,i}^a + (1 - \gamma) S_{FP,i}^a]^{1/(a-1)}$. Long-run percent changes:

SPECIFICATION	NF PRICE	FP PRICE	NF STOCK	FP STOCK	TOTAL
Base model ($\sigma = \infty$)	+0.150%	-28.782%	+0.180%	-33.456%	-0.175%
CES $\sigma = 0.5$	+9.931%	-21.826%	+12.033%	-25.583%	+11.636%
CES $\sigma = 1$	+0.182%	-28.759%	+0.218%	-33.430%	-0.136%
CES $\sigma = 2$	+0.017%	-28.870%	+0.021%	-33.562%	-0.333%
CES $\sigma = 4$	+0.005%	-28.885%	+0.006%	-33.571%	-0.347%

07 · WHERE THE RIVER ENDS

Safer ground, tighter markets.

Risk Rating 2.0 can reduce development in high-risk areas and support climate adaptation.

It can also tighten regional supply and worsen affordability when destination markets cannot expand quickly or elastically enough.

Complementary housing-supply policies may be needed to reduce those affordability costs.

Housing is the principal store of non-wage wealth for many low- and middle-income households. Capitalization losses and higher housing costs may bring financial hardship, forced doubling-up, or housing instability. Those distributional consequences motivate a separate welfare analysis and are not yet quantified.

These simulations are disciplined equilibrium counterfactuals, not causal estimates of Risk Rating 2.0.

Next. Every U.S. county. Estimate α . Trace how flood risk, premium changes, market size, and supply constraints shape adaptation and affordability.

Price the risk. Then make room where the water does not reach.